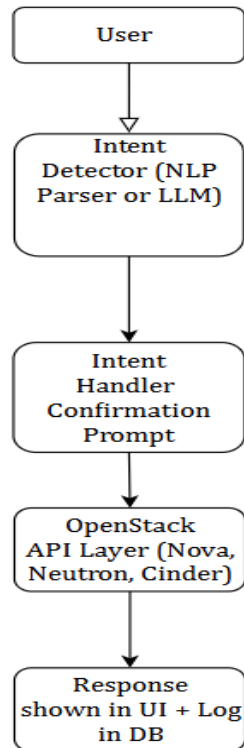


Documentation: Agentic AI for Cloud Operations

1. Architecture Overview :

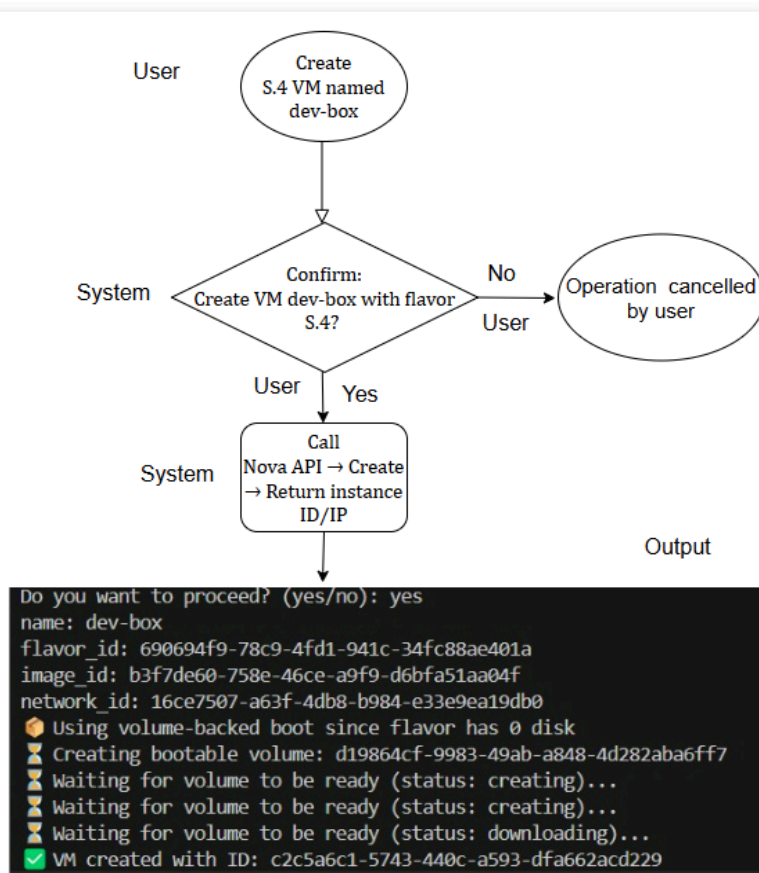


Components:

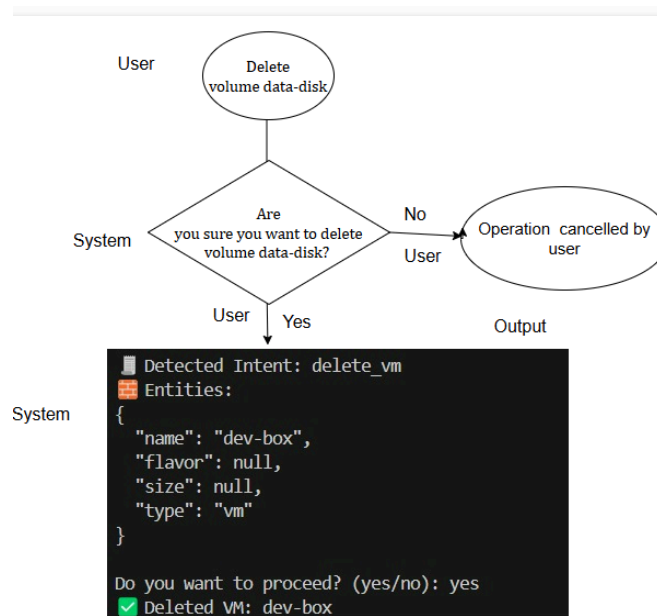
- NLP Parser – Detects intent (create/delete/resize)
- Dialog Manager – Handles confirmation logic
- API Client – Talks to OpenStack using tokens
- Database – Stores all user requests

2.Dialog-Flow Diagrams :

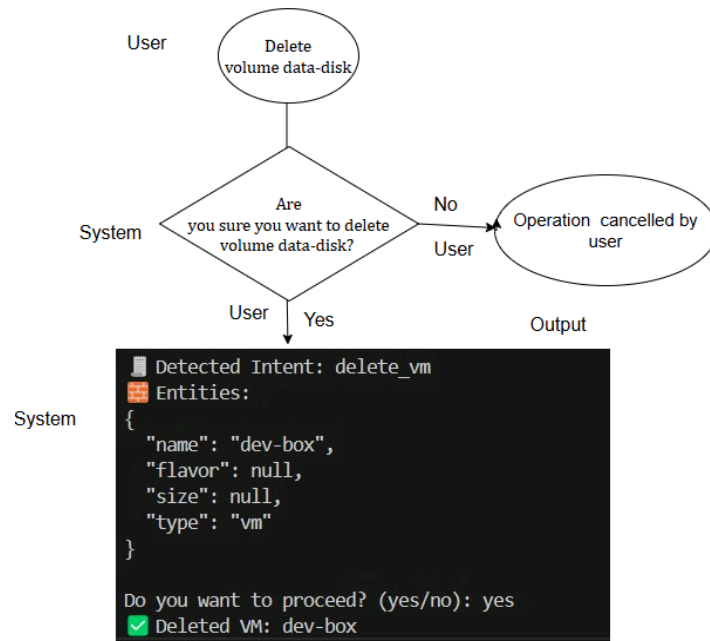
Scenario: VM Provisioning :



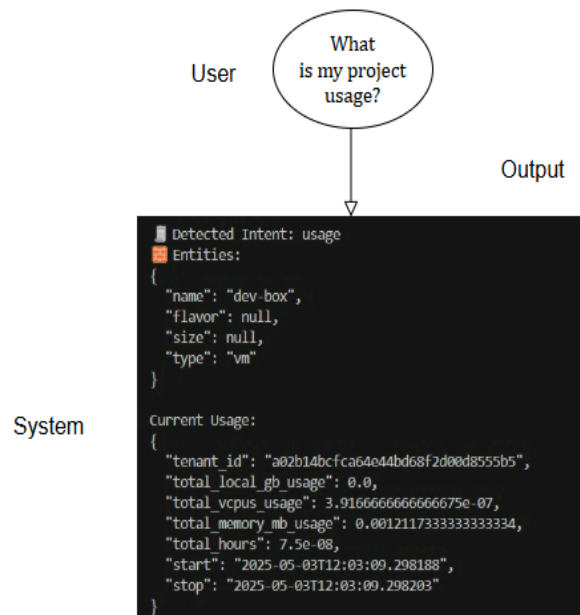
Scenario: VM Deletion :



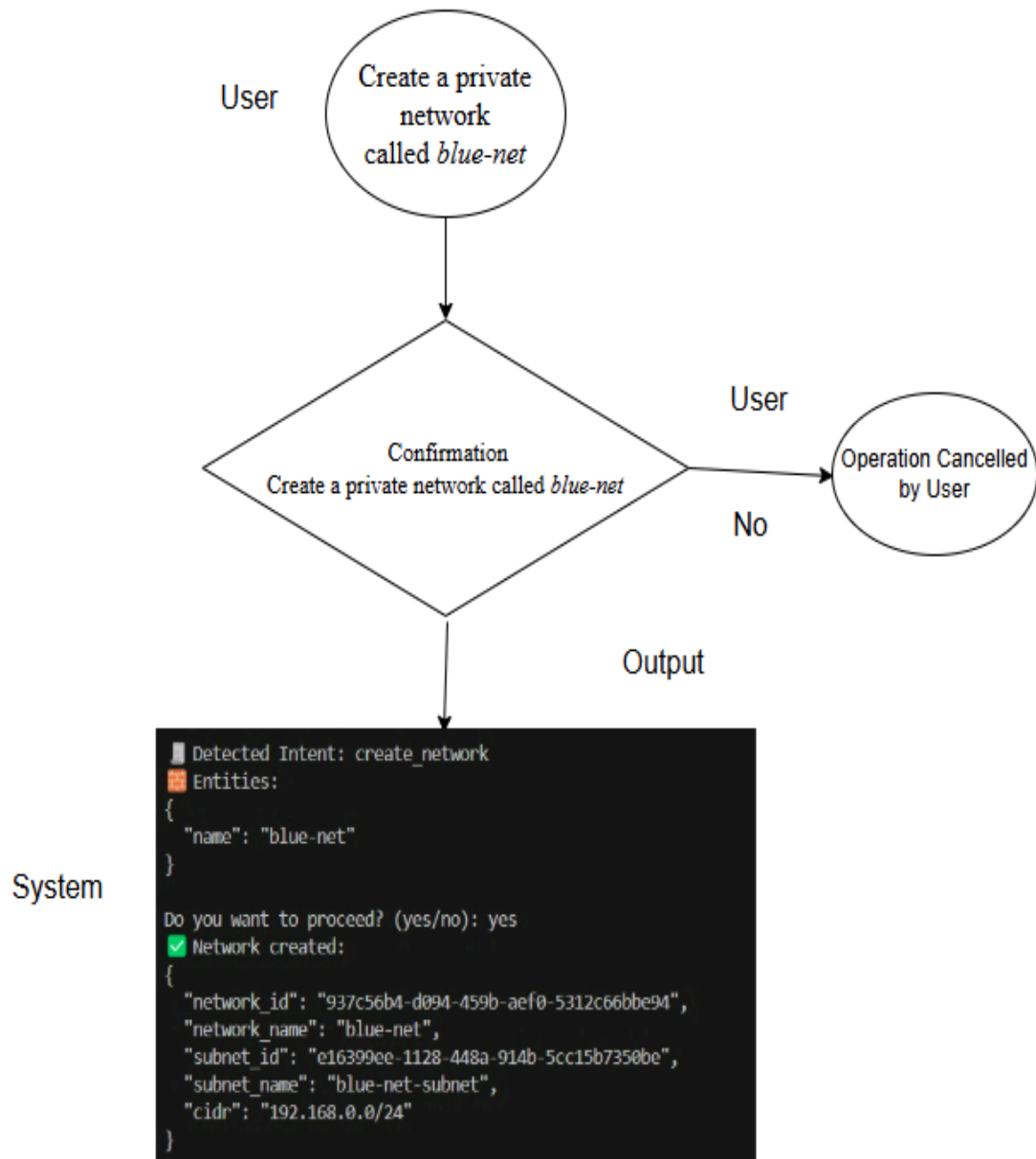
Scenario: Volume Operations :



Scenario: Usage Query :



Scenario: Network Creation :



3. Test-Case Matrix :

Test Case ID	Functional Requirement	Test Scenario	Input Command	Expected Output / Behavior	Pass Criteria
TC-01	FR-1: VM Provisioning	Create a VM with specific flavor and name	Create an S.4 VM named dev-box	Model confirms, launches VM, returns instance ID and IP address	ID & IP returned; VM is visible in list
TC-03	FR-3: VM Deletion	Delete a specific VM	Delete the VM dev-box	Model confirms deletion, removes VM from project	VM no longer exists
TC-05	FR-5: Volume Operations	Create a volume with given name and size	Create a 100 GB volume named data-disk.	Model confirms and creates volume	Volume is listed with correct size & name
TC-06	FR-6: Usage Query	Ask for current project usage	What's my project usage?	Returns vCPU, RAM, GPU, and volume usage in current project	Accurate resource usage is shown
TC-04	FR-4: Network Creation	Create a private network with a name	Create a private network called blue-net	Model confirms, creates network + subnet, returns network/subnet IDs	Network is visible with correct settings